American Computer Science League

2022-2023 ● Contest 3: Short Problem Solutions ● Elementary Division

1. Boolean Algebra

The expression $(x \geq 6)\ AND\ (x < 8)\ OR\ NOT\ (x < 10)$ is TRUE for all integers greater than or equal to 6 and less than 8 which are {6, 7} as well as the integers that are not less than 10 which are {10, 11, 12, 13}. That's 6 of them.

C. 6

2. Boolean Algebra

The Boolean expression $\sim(\sim A\sim B) + \sim(A + B)$ can be simplified to $A + B + \sim A\sim B$. A is TRUE for (1,1) and (1,0). B is TRUE for (0,1) and (1,1). ~A~B is only TRUE for (0,0). All 4 ordered pairs make it TRUE.

E. 4

3. Boolean Algebra

a. $\sim B(A\sim B + A) = A\sim B + A\sim B = A\sim B$
b. $\sim A(B + A) = \sim AB + \sim AA = \sim AB + 0 = \sim AB$
c. $\sim(A + B) = \sim A\sim B$
Therefore, c is equivalent to d which is already in simplified form.

D. c,d

4. Boolean Algebra

The Boolean expression $\sim AB(\sim A + \sim B) + A\sim B(\sim A + \sim B)$ simplifies to $\sim AB + \sim AB\sim B + A\sim B\sim A + A\sim B\sim B$. The second and third terms cancel out so the simplified form is $\sim AB + A\sim B$.

B. $A\sim B + \sim AB$

5. Boolean Algebra

A tautology is always TRUE. The 4 expressions simplify as follows:
a. $A(\sim B + B) + \sim A = A\sim B + AB + \sim A = A(\sim B + B) + \sim A$
$= A(1) + \sim A = A + \sim A = 1$
b. $\sim(A + B) + (A + B) = \sim A\sim B + A + B = 1$
c. $A(A\sim B) + B(\sim AB) = AA\sim B + \sim ABB = A\sim B + \sim AB$
d. $AB + \sim AB + \sim B = (A + \sim A)B + \sim B = (1)B + \sim B = B + \sim B$
$= B + \sim B = 1$
All but c simplifies to 1 which makes the expression always TRUE in a, b, and d.

C. c